WEEK 3 – DATA SCIENCE AND VISUALIZATION

The Cleaning Sprint: Missing Values and Outlier Detection

Project: Accident Data Analysis

Dataset: AccidentsBig.csv

1. Introduction

Week 3 focuses on the data-cleaning stage of the accident-data project. The raw dataset is first examined to establish a baseline, after which missing values and potential outliers are investigated. The cleaned dataset is then compared with the original dataset to demonstrate the effect of the cleaning process.

2. Objectives

The objectives are to assess data quality before cleaning, identify and quantify missing values, handle missing numerical and categorical values using appropriate imputation strategies, inspect duplicate records, detect potential outliers using boxplots and Z-score analysis, and compare the dataset before and after cleaning.

3. Dataset Used

The analysis uses the **AccidentsBig.csv** dataset, containing accident-related attributes with a mixture of numerical and categorical information. The same raw dataset used during Week 2 is used as the starting point for Week 3.

4. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, and Jupyter Notebook are used for the data-cleaning and quality-assessment process.

5. Raw Dataset – Before Cleaning

The original dataset is loaded without modifying its values. A working copy is created so the raw data remains available for comparison.

```
df_raw = pd.read_csv("AccidentsBig.csv", low_memory=False)
df = df_raw.copy()
print("Shape before cleaning:", df.shape)
```

6. Missing-Value Analysis – Before Cleaning

Missing values are identified and quantified before any values are changed. This establishes the baseline for the before-and-after comparison.

```
missing_before = df.isnull().sum()
missing_before = missing_before[missing_before > 0].sort_values(ascending=False)
missing_before_pct = (df.isnull().sum() / len(df)) * 100
missing_before_summary = pd.DataFrame({
"Missing Values": df.isnull().sum(),
"Missing Percentage": missing_before_pct.round(2)
}).query("`Missing Values` > 0").sort_values("Missing Values", ascending=False)
missing_before_summary
```

7. Duplicate Analysis – Before Cleaning

Duplicate records are counted before cleaning so that the original dataset provides a baseline for later comparison.

```
duplicates_before = df.duplicated().sum()
print("Duplicate records before cleaning:", duplicates_before)
```

8. Descriptive Statistics – Before Cleaning

Descriptive statistics provide a baseline for the numerical variables before cleaning.

```
df.describe().T
```

9. Missing-Value Visualization – Before Cleaning

A bar chart visually represents the number of missing values in affected columns.

```
missing_plot = df.isnull().sum()
missing_plot = missing_plot[missing_plot > 0].sort_values(ascending=False)
plt.figure(figsize=(12, 6))
if len(missing_plot) > 0:
sns.barplot(x=missing_plot.values, y=missing_plot.index)
plt.title("Missing Values Before Cleaning")
plt.xlabel("Number of Missing Values")
plt.ylabel("Column")
plt.show()
```

10. Missing-Value Handling

Numerical missing values are handled using median imputation, while categorical missing values are handled using mode imputation. Median is less sensitive to extreme values than the mean, while mode is appropriate for categorical data.

10.1 Numerical Columns – Median Imputation

```
numerical_columns = df.select_dtypes(include=np.number).columns.tolist()
for col in numerical_columns:
if df[col].isnull().sum() > 0:
df[col] = df[col].fillna(df[col].median())
```

10.2 Categorical Columns – Mode Imputation

```
categorical_columns = df.select_dtypes(include=["object", "category"]).columns.tolist()
for col in categorical_columns:
if df[col].isnull().sum() > 0:
mode_value = df[col].mode(dropna=True)
if not mode_value.empty:
df[col] = df[col].fillna(mode_value.iloc[0])
```

11. Missing Values – After Imputation

The dataset is checked again after imputation to determine whether missing values remain.

```
missing_after_imputation = df.isnull().sum()
missing_after_imputation = missing_after_imputation[missing_after_imputation > 0]
if len(missing_after_imputation) == 0:
print("No missing values remain after imputation.")
else:
print(missing_after_imputation)
```

12. Outlier Detection Using Boxplots

Boxplots identify potential observations outside the normal whisker range. Potential outliers are not automatically removed because unusual accident records may be legitimate.

```
plot_columns = numerical_columns[:6]
for col in plot_columns:
plt.figure(figsize=(8, 4))
sns.boxplot(x=df[col].dropna())
plt.title(f"Boxplot of {col}")
plt.xlabel(col)
plt.show()
```

13. Outlier Detection Using Z-score

Z-score analysis provides a quantitative method for identifying observations far from the mean. The commonly used threshold $|Z| > 3$ is used to flag potential outliers.

```
z_scores = df[numerical_columns].apply(zscore, nan_policy="omit")
outlier_counts = (z_scores.abs() > 3).sum()
outlier_percentages = (outlier_counts / len(df)) * 100
outlier_summary = pd.DataFrame({
"Potential Outliers": outlier_counts,
"Percentage": outlier_percentages.round(2)
}).sort_values("Potential Outliers", ascending=False)
outlier_summary
```

14. After Cleaning – Dataset Quality

After the cleaning operations, the dataset is checked again for missing values, duplicate records, dimensions, and descriptive statistics.

```
print("Shape after cleaning:", df.shape)
print("Total missing values:", df.isnull().sum().sum())
print("Duplicate records after cleaning:", df.duplicated().sum())
```

15. Before vs After – Missing Values

The total number of missing values before and after cleaning is compared to measure the effect of imputation.

```
comparison_missing = pd.DataFrame({
"Stage": ["Before Cleaning", "After Cleaning"],
"Total Missing Values": [df_raw.isnull().sum().sum(), df.isnull().sum().sum()]
})
comparison_missing
```

16. Before vs After – Duplicate Records

```
comparison_duplicates = pd.DataFrame({
"Stage": ["Before Cleaning", "After Cleaning"],
"Duplicate Records": [duplicates_before, df.duplicated().sum()]
})
comparison_duplicates
```

17. Before vs After – Dataset Dimensions

```
comparison_shape = pd.DataFrame({
"Stage": ["Before Cleaning", "After Cleaning"],
"Rows": [df_raw.shape[0], df.shape[0]],
"Columns": [df_raw.shape[1], df.shape[1]]
})
comparison_shape
```

18. Before vs After – Missing-Value Visualization

A final comparison plot provides a visual representation of how missing-value counts changed after cleaning.

```
before_counts = df_raw.isnull().sum()
after_counts = df.isnull().sum()
comparison = pd.DataFrame({"Before Cleaning": before_counts, "After Cleaning": after_counts})
comparison = comparison[(comparison["Before Cleaning"] > 0) | (comparison["After Cleaning"] > 0)]
comparison.plot(kind="bar", figsize=(14, 6))
plt.title("Missing Values: Before vs After Cleaning")
plt.xlabel("Columns")
plt.ylabel("Number of Missing Values")
plt.xticks(rotation=75)
plt.legend()
plt.tight_layout()
plt.show()
```

19. Cleaned Dataset

The cleaned dataset is saved separately so the original raw dataset remains unchanged and the cleaned data can be used in later project stages.

```
output_file = "AccidentsBig_cleaned_week3.csv"
df.to_csv(output_file, index=False)
print(f"Cleaned dataset saved as: {output_file}")
```

20. Final Verification

The final verification confirms the shape, total missing values, and duplicate count of the cleaned dataset.

```
print("Final shape:", df.shape)
print("Total missing values:", df.isnull().sum().sum())
print("Duplicate records:", df.duplicated().sum())
```

21. Conclusion

Week 3 focused on systematic cleaning and quality assessment of the accident dataset. The raw dataset was first analyzed to establish a before-cleaning baseline. Missing values were handled using median imputation for numerical variables and mode imputation for categorical variables. Potential outliers were investigated through boxplots and Z-score analysis. Finally, the cleaned dataset was compared with the raw dataset using missing-value counts, duplicate counts, dataset dimensions, and visual comparison. The cleaned dataset was saved separately for subsequent project stages.

22. Project Status

Week 3: Missing Value Handling, Outlier Detection, and Before-vs-After Cleaning Analysis — Completed